



Figure 4: Extracting features from the spectrogram

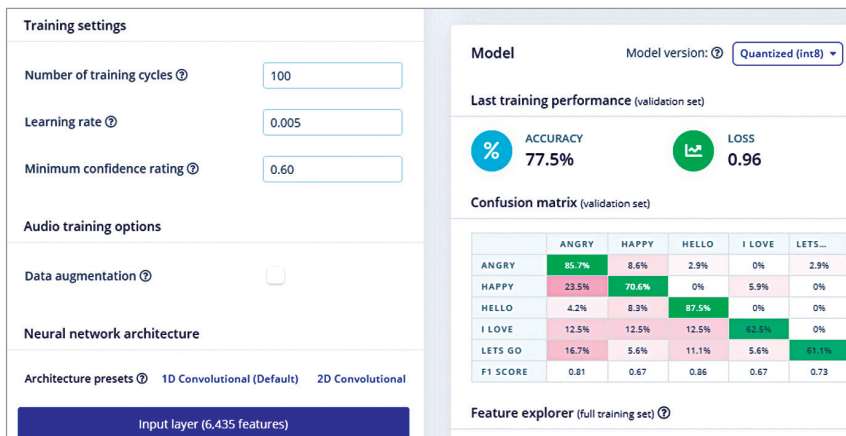


Figure 5: Accuracy output during ML model testing

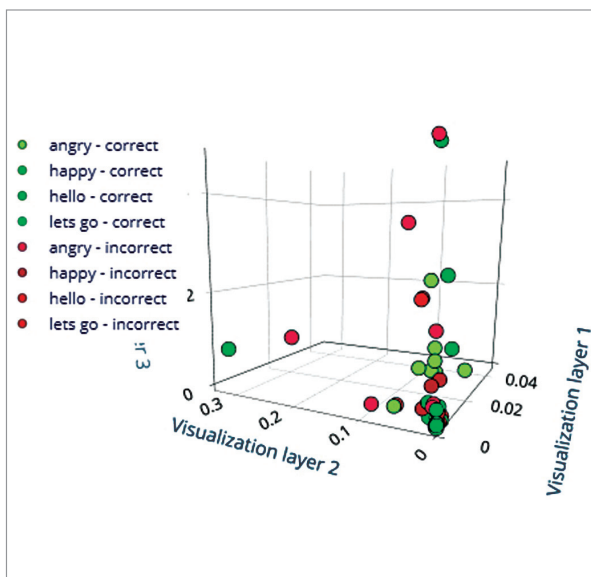


Figure 6: 3D Graph showing correct/incorrect recognition during testing of ML model

```
sudo apt-get install libatlas-base-dev
libportaudio0 libportaudio2 libportaudi
ocpp0 portaudio19-dev
pip3 install edge_impulse_linux -i
https://pypi.python.org/simple
pip3 install edge_impulse_linux
git clone https://github.com/
edgeimpulse/
linux-sdk-python
```

Coding

Create a .py file called *animal_translate* and import *espeak* in the Python code, which translates the animal sounds so that humans can understand them. Create an *if* condition for checking the accuracy of the emotion present in the output of the ML model. If the output accuracy of 98% or more for a particular label is detected, the output matches the label description. For example, if an animal's sound for the label "hello" meets the said percentage, the animal is indeed saying "hello."

Testing

Download the .iem ML model file, open the terminal, and run *animal_translate.py* followed by the path location of the ML model. Select an animal, say elephant. So, whenever

```
import os
import sys, getopt
import signal
import time
from espeak import espeak
from edge_impulse_linux.audio import AudioImpulseRunner

runner = None

def signal_handler(sig, frame):
    print('Interrupted')
    if (runner):
        runner.stop()
    sys.exit(0)

signal.signal(signal.SIGINT, signal_handler)

def help():
    print('python classify.py <path_to_model.iem> <audio_device_ID, optional>')

def main(argv):
    try:
        opts, args = getopt.getopt(argv, "h", ["--help"])
    except getopt.GetoptError:
        help()
        sys.exit(2)
    for opt, arg in opts:
        if opt in ('-h', '--help'):
            help()
            sys.exit(2)

    if len(args) == 0:
        help()
        sys.exit(2)

    model = args[0]

    dir_path = os.path.dirname(os.path.realpath(__file__))
    modelfile = os.path.join(dir_path, model)
```

Figure 7: Python code